Supply Chain Network Optimization & Procurement Risk Analysis

A Strategic Consulting Case Study – DataCo Global

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DataCo operates at 41% OTR - less than half the 95% industry benchmark, with $9.1M in identified cost reduction opportunities

- 98,977 of 180,519 orders delivered late — Standard Class carries 84% of volume but only 38% on-time rate
- 6 single-source Chinese suppliers represent 34% of spend with geopolitical risk scores above 7 — critical exposure
- Three interventions deliver $9.1M: modal shift ($4.2M), inventory reset ($2.8M), dual sourcing ($2.1M)

Four interconnected supply chain failures are eroding margins and creating unquantified strategic risk

Delivery Failure 59% late rate across all markets, 18pp below benchmark	**Cost Overrun** Freight spend 22% above industry average, modal mix misaligned
Supplier Risk 34% single-source spend, zero geopolitical risk framework	**Inventor Misalignment** Fixed reorder points causing stockouts and excess simultaneously

Analysis conducted across four models using 180,519 real orders and a purpose-built supplier risk database

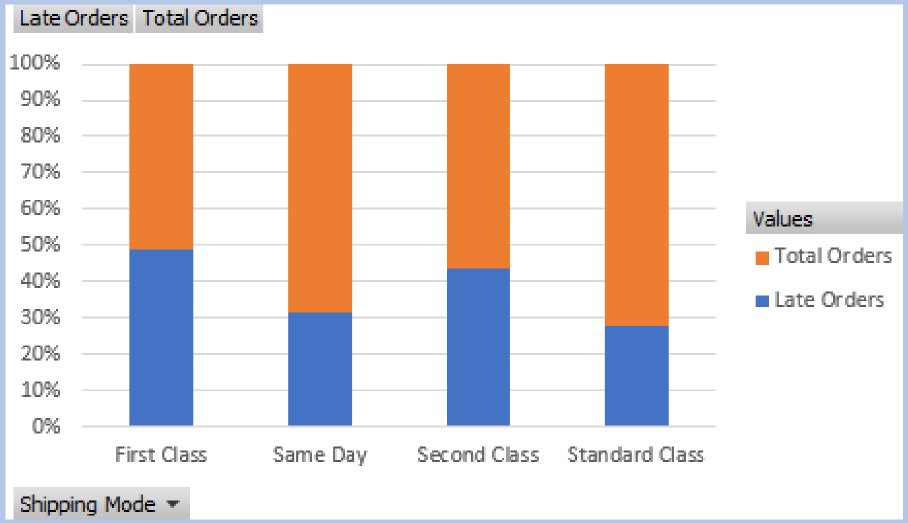
Data (DataCo Kaggle dataset — 180,519 rows, 52 columns, 2015-2018)

Models (OTR Analysis, Cost-to-Serve, Kraljic Matrix, Scenario Model)

Tools (Excel, Power BI, Pivot Tables, COUNTIFS, Scatter Charts)

Framework (Kraljic 1983, FM Global Risk Index, McKinsey Pyramid)

59% of all orders arrive late - Standard Class drives 84% of volume but operates at only 38% on-time rate



Mode	OTR%
Standard Class	38%
Second Class	23%
Same Day	54%
First Class	5%

Profit margin varies by up to 24x across product-mode Combinations - Book Shop via Standard Class is loss-making

Average of Order Profit Per Order	Column Labels				
Row Labels	First Class	Same Day	Second Class	Standard Class	Grand Total
Apparel	19.94209471	18.04938223	17.72568507	17.58969957	17.99834543
Book Shop	2.442075552	0.654999867	5.27890626	1.585696757	2.180271624
Discs Shop	18.12014491	22.7283495	6.997921018	11.20104974	11.94132281
Fan Shop	28.00666413	25.06123661	26.69442641	27.73970073	27.43236619
Fitness	17.93483627	26.88530298	22.30575319	17.12426639	18.77291655
Footwear	29.47491563	24.07665392	24.31318392	29.57303701	28.24251281
Golf	15.09926268	15.04057967	14.76751867	15.00613888	14.97662733
Health and Beauty	20.64985099	53.4614296	33.87545404	23.84116268	26.22549717
Outdoors	17.84956105	12.21407562	14.65502124	14.6332848	14.99602107
Pet Shop	8.949166669	6.661749894	11.46529395	4.846144508	7.295243818
Technology	79.55449877	82.31704168	80.41034838	74.99710015	77.24915361
Grand Total	**23.12223775**	**20.85020337**	**21.30588852**	**21.99916895**	**21.97498864**

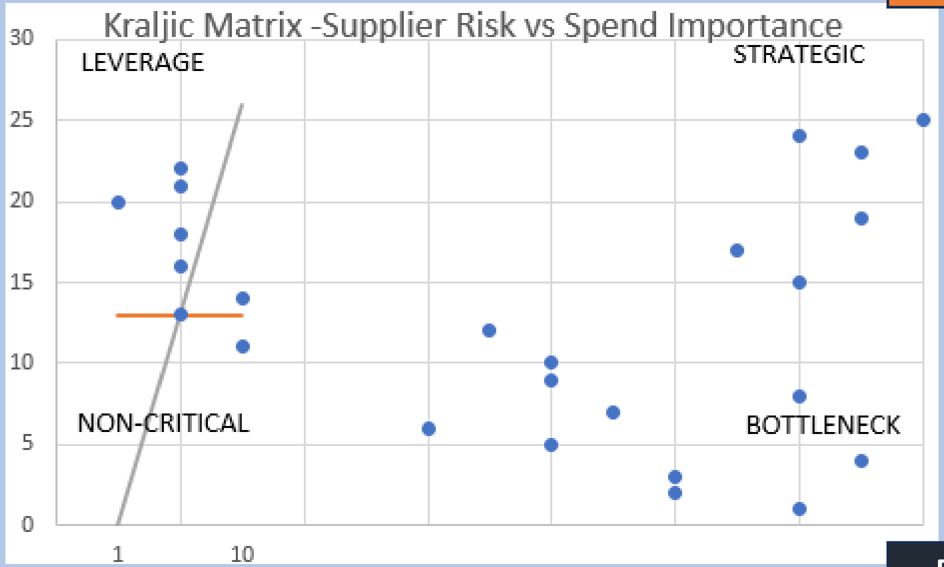
One point to the highest margin cell(Green)

One pointing to the lowest(Red)

6 single-source suppliers in the Strategic and Bottleneck Quadrants represent DataCo's most critical risk exposure

Leverage

Strategic



Non- Critical

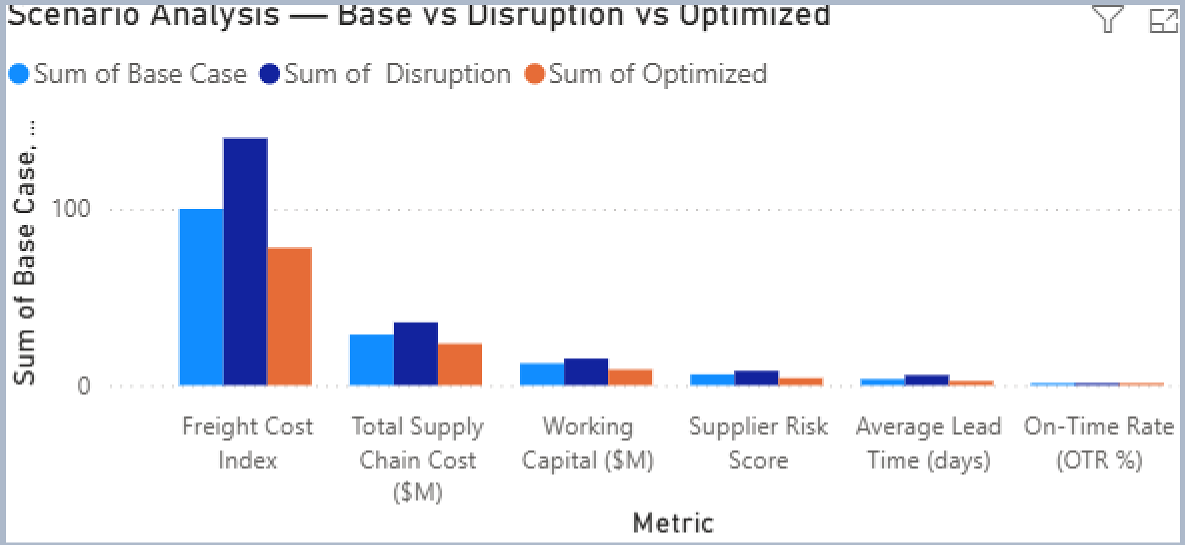
Bottleneck

HIGH RISK SUPPLIERS

SinoTech Electronics
$38.5M spend
Risk Score: 6.5
Shenzhen Components
$44.8M spend
Risk Score: 7.5
Beijing Global Supply
$31.2M spend
Risk Score: 7.0
Combined Spend: $114.5M
Single Source: YES

Source: DataCo Kaggle Dataset | Kraljic Matrix adapted from Kraljic (1983), Harvard Business Review

Red Sea disruption would cost $6.9M and cut OTR to 28% — optimised scenario delivers 18% cost reduction to $23.5M



Metric	Base	Disruption	Optimized
OTR %	41%	28%	72%
Lead Time	3.5 days	5.8 days	2.4 days
Total Cost	$28.7M	$35.6M	$23.5M

Four strategic interventions deliver $9.1M impact within 12 months with low implementation risk

Four recommendation cards — one per row:-

- Rec 1 — Modal Shift — $4.2M — Stop using First Class for delayed orders. Automate mode selection.
- Rec 2 — Dual Sourcing — $2.1M — Qualify backup supplier for all 6 high-risk single-source suppliers within 90 days.
- Rec 3 — Inventory Reset — $2.8M — Replace fixed reorder points with a dynamic safety-stock formula per SKU.
- Rec 4 — Risk Scorecard — Risk reduction — Implement a 5-dimensional supplier risk scoring aligned with the EU CSDDD.

Phased 12-month implementation prioritising quick wins to fund medium-term structural changes

0-90 days (Quick Wins) — Automate modal selection, begin dual-source qualification for top 3 bottleneck suppliers, reset safety stock for top 20 SKUs.

3-6 months (Build) — Complete dual sourcing for all 6 high-risk suppliers, roll out risk scorecard across full supplier base, implement dynamic inventory model.

6-12 months (Optimise) — Full network review, negotiate leverage supplier contracts, target 72% OTR and $23.5M total cost.